

Software Quality Engineering

Static Analysis, Human Review & Quality Improvement Report

Python Quiz App — SQE Assignment

Repository: jordanonlycodes/SQE-Quiz-App

Branch: static-analysis

Baseline source commit: ac648a2

Baseline Pylint commit: 91c1f2e

Improvement commits: f25825a, 2c5022d, ea96954, 5449e69

Executive Summary

The project is a PySide6 desktop quiz application that retrieves questions from the Open Trivia Database API, loads them through a worker-thread architecture, presents the quiz through a GUI, and includes automated tests. The baseline Pylint run produced 695 findings and a score of 4.49/10. The analysis deliberately distinguishes static-analysis findings from confirmed defects: 303 findings are concentrated in protected-access/test patterns and fixture naming, while 178 are formatting findings. Human review was therefore used to identify actionable design and maintainability problems instead of treating every Pylint message as a defect.

The implemented improvements focus on object initialization, control-flow clarity, encoding robustness, naming, duplication reduction, and decomposition of question-data transformation. The framework-independent backend regression suite passes 19/19. The full GUI pytest suite and a new final Pylint execution could not be run in this environment because PySide6, pytest-mock, pytest-qt, and Pylint are unavailable and external package installation is unavailable.

1. Project Selection & Code Statistics

The project satisfies the assignment selection criteria: it is primarily Python, contains multiple modules/classes, has substantially more than 800 source lines, integrates an external API and GUI framework, and contains an established automated test suite.

Item	Evidence
Project	Quiz App
Repository	jordanonlycodes/SQE-Quiz-App
Purpose	Desktop quiz application using Python, PySide6 and OpenTDB API
Source Python LOC	1,454
Test Python LOC	2,495
Python files	39 source/test files
Tests	pytest/pytest-qt + unittest
Baseline Pylint	4.49/10; 695 findings
Working branch	static-analysis

Baseline preservation

The original source is preserved in Git at commit `ac648a2`. The initial Pylint output is preserved in `deliverable/pylint_baseline.txt` and `pylint_baseline.json`. Improvement work begins after baseline analysis commit `91c1f2e`.

2. Initial Pylint Baseline

Metric	Baseline
Overall score	4.49 / 10
Total findings	695
Warnings	323
Conventions	256
Errors	90
Refactoring	26

$W0212 + W0621 = 303$ findings. $C0303 + C0301 + C0304 = 178$ findings. Together these account for $481/695 = 69.2\%$ of all findings. This shows that the raw count substantially overstates the number of confirmed software defects.

3. Classification of Findings

Category	Main messages	Count	Assessment
Test/static-analysis interaction	W0212, W0621	303	Mostly test conventions/internal inspection; context dependent.
Formatting/style	C0303, C0301, C0304	178	Real consistency concerns, but low severity.
Dependency/environment	E0401, E0611	89	Requires environment-aware interpretation; PySide6/pytest analysis limitations.
Documentation	C0114, C0115, C0116	63	Maintainability issue; not every missing docstring is equally important.
Design/maintainability	R0903, R0902, R0913, R0917, R2020, R0801	26	Highest-value area for human review.
Other correctness/style	W0201, W0718, C0123, W1514, W0622, etc.	30	Mixed; investigate individually.

The classification is intentionally quality-oriented rather than simply reproducing Pylint's C/W/R/E labels.

1. W0201 — attribute-defined-outside-init

Location: modules/gui/main_window.py

Human decision: A — Should be fixed

Original code

```
25 |  
26 |     def __init__(self) -> None:  
27 |         super().__init__()  
28 |         self._setup_window()  
29 |         self._setup_layout()  
30 |         self._init_default_variables()  
31 |         self._setup_menu_bar()  
32 |         self._setup_error_overlay()  
33 |         self._setup_loading_overlay()  
34 |         self._setup_initial_widget()  
35 |  
36 |     def _setup_window(self) -> None:  
37 |         self.setObjectName("mainWindow")  
38 |         self.setWindowTitle('Quiz')  
39 |         self.resize(896, 512)  
40 |  
41 |     def _setup_layout(self) -> None:  
42 |         self._main_layout = QStackedLayout()  
43 |         self._main_layout.setStackingMode(QStackedLayout.StackingMode.StackAll)  
44 |         self._central_widget = QWidget()  
45 |         self._central_widget.setLayout(self._main_layout)  
46 |         self.setCentralWidget(self._central_widget)  
47 |  
48 |     def _init_default_variables(self) -> None:  
49 |         # List of ignored widgets, from MainWindow clean method  
50 |         # Used for overlays  
51 |         self._ignored = []  
52 |         self._close_requested = False
```

What the problem means: MainWindow created important state in a separate initialization helper and later methods. This made the valid object state less obvious and caused repeated assignment warnings.

Quality impact: Maintainability, readability, and testability are affected because a maintainer can see the complete initial state in one place.

Decision rationale/status: Fixed by moving the state declarations into `__init__`.

2. W0718 — broad-exception-caught

Location: modules/gui/workers/question_loader.py / main_window.py / worker_thread_controller.py

Human decision: B — Context dependent

Original code

```
35 |         try:
36 |             logger.info(
37 |                 "QuestionLoader started: amount=%s, category=%s, difficulty=%s, type=%s",
38 |                 self._amount,
39 |                 self._category or "any",
40 |                 self._difficulty or "any",
41 |                 self._question_type or "any",
42 |             )
43 |             questions = self._load_questions()
44 |             logger.info("QuestionLoader loaded %s questions.", len(questions.questions_list))
45 |             self.loaded.emit(questions)
46 |
47 |         except OpenTriviaClientError as error:
48 |             logger.warning("Question loading failed: %s: %s", type(error).__name__, error, exc_info=True)
49 |             self.error.emit(error)
50 |
51 |         except Exception as error:
52 |             logger.exception("Unexpected question loading error.")
53 |             self.error.emit(error)
54 |
55 |         finally:
56 |             self.finished.emit()
57 |
```

What the problem means: Catching Exception is broad and can hide programming defects. However, these catches occur at GUI/worker boundaries where unexpected failures need to be logged and converted into signals instead of killing the GUI.

Quality impact: Reliability is relevant, while maintainability/debuggability can suffer if the boundary is used indiscriminately.

Decision rationale/status: Retained as context dependent.

3. R1720 — no-else-raise

Location: modules/questions/opentdb_client.py

Human decision: A — Should be fixed

Original code

```
77 |         try:
78 |             response = requests.get(self.BASE_URL, params=params, timeout=3)
79 |             logger.debug("OpenTDB response status: %s", response.status_code)
80 |             response.raise_for_status()
81 |             data = response.json()
82 |             if not data['results']:
83 |                 raise NoQuestionsFoundError('No questions found for selected parameters. Try different category, difficulty or type')
84 |             elif int(amount) > len(data['results']):
85 |                 raise NotEnoughQuestionsError('Not enough questions found for the selected parameters. Try lower number.')
86 |             logger.debug(
87 |                 "OpenTDB response parsed: response_code=%s, results=%s",
```

What the problem means: The first branch raises an exception, so the following elif does not need to be attached to it.

Quality impact: Readability and maintainability improve because the two validation conditions are visibly independent.

Decision rationale/status: Fixed by changing the elif after raise to a separate if.

4. R0913 — too-many-arguments

Location: modules/questions/models.py

Human decision: B — Context dependent

Original code

```
10 | class Question:
11 |     """Data object for one quiz question and its scoring data."""
12 |
13 |     def __init__(
14 |         self, tp: str,
15 |         difficulty: str,
16 |         category: str,
17 |         question: str,
18 |         correct_answer: str,
19 |         incorrect_answers: list[str],
20 |         all_answers: list[str],
21 |         points: int,
22 |     ) -> None:
23 |         self.tp = tp
24 |         self.difficulty = difficulty
25 |         self.category = category
26 |         self.question = question
27 |         self.correct_answer = correct_answer
28 |         self.incorrect_answers = incorrect_answers
```

What the problem means: Question is a data object and the constructor arguments correspond directly to the state needed to represent one question.

Quality impact: A large constructor can reduce readability, but adding a builder/factory solely to lower the count could add unnecessary abstraction.

Decision rationale/status: Kept as context dependent.

5. R0917 — too-many-positional-arguments

Location: modules/questions/models.py

Human decision: B — Context dependent

Original code

```
10 | class Question:
11 |     """Data object for one quiz question and its scoring data."""
12 |
13 |     def __init__(
14 |         self, tp: str,
15 |         difficulty: str,
16 |         category: str,
17 |         question: str,
18 |         correct_answer: str,
19 |         incorrect_answers: list[str],
20 |         all_answers: list[str],
21 |         points: int,
22 |     ) -> None:
23 |         self.tp = tp
24 |         self.difficulty = difficulty
25 |         self.category = category
26 |         self.question = question
27 |         self.correct_answer = correct_answer
28 |         self.incorrect_answers = incorrect_answers
```

What the problem means: Eight positional values are passed to Question. This is less self-documenting than keywords, but the object is simple and directly represents those fields.

Quality impact: Readability and modifiability may be affected if fields change frequently.

Decision rationale/status: Kept as context dependent.

6. R0902 — too-many-instance-attributes

Location: modules/gui/main_window.py

Human decision: B — Context dependent

Original code

```
14 | class MainWindow(QMainWindow):
15 |     """
16 |     Main application window and top-level GUI coordinator.
17 |
18 |     MainWindow owns the central stacked layout, menu bar, loading/error overlays,
19 |     and the question-loading worker controller. It coordinates navigation between
20 |     the welcome screen, start screen, and question display.
21 |
22 |     Question loading is delegated to a worker thread. The OpenTDB request timeout
23 |     is configured in OpenTriviaClient.
24 |     """
25 |
26 |     def __init__(self) -> None:
27 |         super().__init__()
28 |         self._setup_window()
29 |         self._setup_layout()
30 |         self._init_default_variables()
31 |         self._setup_menu_bar()
32 |         self._setup_error_overlay()
33 |         self._setup_loading_overlay()
34 |         self._setup_initial_widget()
```

What the problem means: MainWindow owns GUI state and coordinates application-level navigation. GUI coordinator classes naturally have more references than small value objects.

Quality impact: The count can signal low cohesion, but it is not proof of poor design.

Decision rationale/status: Kept as context dependent; architecture is better judged by responsibility than by an arbitrary threshold.

7. R0801 — duplicate-code

Location: tests/unittest_backend/test_opentdb_client_unittest.py

Human decision: B — Context dependent

Original code

```
70 |  
71 |         mock_requests_get.assert_called_once_with(  
72 |             OpenTriviaClient.BASE_URL,  
73 |             params={  
74 |                 "amount": 2,  
75 |                 "category": "9",  
76 |                 "difficulty": "easy",  
77 |                 "type": "boolean",  
78 |             },  
79 |             timeout=3,  
80 |         )  
81 |  
82 |         mock_response.raise_for_status.assert_called_once()  
83 |  
84 |         @patch("modules.questions.opentdb_client.requests.get")  
85 |         def test_get_questions_data_raises_error_when_no_questions(self, mock_requests_get: MagicMock) -> None:  
86 |             """Test that an empty results list raises OpenTriviaClientError."""  
87 |             mock_response = Mock()  
88 |             mock_response.status_code = 200  
89 |             mock_response.json.return_value = {  
90 |                 "response_code": 0,
```

What the problem means: The project intentionally contains both unittest and pytest backends, so similar setup and assertions are partly a consequence of maintaining two test styles.

Quality impact: Duplication increases maintenance cost, but deleting one suite could remove intentional coverage/style comparison.

Decision rationale/status: Kept as context dependent.

8. W1514 — unspecified-encoding

Location: main.py

Human decision: A — Should be fixed

Original code

```
98 |         app.setFont(font)
99 |
100 |         # Load and apply styles from styles.css to the entire application
101 |         stylesheet_path = "modules/gui/styles/styles.css"
102 |         with open(stylesheet_path, "r") as file:
103 |             app.setStyleSheet(file.read())
104 |         logger.debug("App stylesheet loaded from: %s", stylesheet_path)
105 |
```

What the problem means: Opening a text file without specifying an encoding makes behavior depend on the platform default.

Quality impact: Portability and reliability can be affected across systems with different defaults.

Decision rationale/status: Fixed by explicitly using UTF-8.

9. C0123 — unidiomatic-typecheck

Location: modules/gui/widgets/overlays/error_overlay.py

Human decision: A — Should be fixed

Original code

```
53 |  
54 |     Args:  
55 |         error: Exception whose message should be shown to the user.  
56 |         for_seconds: Number of seconds before the overlay is hidden.  
57 |  
58 |     Notes:  
59 |         Generic OpenTriviaClientError messages get an additional retry hint.  
60 |     """  
61 |     label_text = str(error)  
62 |     if type(error) is OpenTriviaClientError:  
63 |         label_text += " Try again in a few seconds."  
64 |     self._error_label.setText(label_text)  
65 |     self._error_label.update()  
66 |     self.raise_()  
67 |     self.show()  
68 |     self._error_label.show()
```

What the problem means: `type(error) is OpenTriviaClientError` only matches the exact class. `isinstance()` expresses the type relationship and handles subclasses.

Quality impact: Maintainability and extensibility improve.

Decision rationale/status: Fixed with `isinstance()`.

10. W0622 — redefined-builtin

Location: modules/gui/menu_bar/menus/quiz_menu.py

Human decision: A — Should be fixed

Original code

```
30 |         self.addAction(start_quiz)
31 |
32 |     @Slot()
33 |     def _on_start_quiz_action_triggered(self) -> None:
34 |         self.action_requested.emit(MenuActions.SHOW_START_DISPLAY)
35 |         logger.debug("Start quiz action triggered.")
36 |
37 |     def _add_exit_action(self) -> None:
38 |         exit = QAction('Exit', self)
39 |         exit.setShortcut('Ctrl+Q')
40 |         exit.triggered.connect(self._on_exit_action_triggered)
41 |         self.addAction(exit)
42 |
43 |     @Slot()
```

What the problem means: The local variable `exit` shadows Python's built-in `exit` helper.

Quality impact: Naming clarity improves and accidental confusion with the built-in is avoided.

Decision rationale/status: Fixed by renaming the action to `exit_action`.

4. Manual Code Review — 5 Independent Issues

Human review identified design concerns beyond simply accepting Pylint's message list. At least two are explicitly marked as not directly reported by Pylint.

Manual Issue 1 — Question transformation has multiple responsibilities

Location: modules/questions/models.py

Pylint relationship: Not directly reported by Pylint

Evidence: The original `_questions_data_to_question_objects` method extracted fields, decoded HTML, created answer lists, randomized choices, calculated points, constructed Question objects, and appended them to application state.

Quality impact: Mixed transformation, business rules and object construction reduce cohesion and make focused tests harder.

Recommendation/status: Decomposed into `_build_question`, `_build_answers` and `_get_points`. Implemented in f25825a.

Manual Issue 2 — Repeated parameter-frame construction

Location: modules/gui/widgets/start_display/components/question_params_widget.py

Pylint relationship: Not directly reported by Pylint

Evidence: Four blocks performed the same sequence: construct ParamFrame, append it, add a combo box, and add it to the layout.

Quality impact: Changing common setup required editing multiple near-identical blocks.

Recommendation/status: Extracted `_add_parameter_frame()` while preserving named frame attributes. Implemented in f25825a.

Manual Issue 3 — MainWindow is a high-level coordinator with broad responsibilities

Location: modules/gui/main_window.py

Pylint relationship: Related to R0902, but the responsibility analysis is human review

Evidence: The class coordinates window setup, menu actions, screen transitions, worker lifecycle, loading/error overlays, cleanup and shutdown.

Quality impact: This increases coupling and makes the class a central change point.

Recommendation/status: Keep the coordinator role for now; consider future decomposition based on responsibility, not the R0902 threshold.

Manual Issue 4 — Application startup performs several infrastructure tasks

Location: main.py

Pylint relationship: Not directly reported by Pylint

Evidence: Startup configures the working directory, logging, log cleanup, QApplication, font, stylesheet loading and the main window.

Quality impact: Combining infrastructure setup and application construction makes isolated testing of startup behavior harder.

Recommendation/status: Future improvement: move infrastructure setup behind dedicated bootstrap/services.

Manual Issue 5 — Two test frameworks create deliberate duplication

Location: tests/pytest_backend and tests/unittest_backend

Pylint relationship: Partly reflected by R0801

Evidence: The same backend behavior is tested in pytest and unittest styles.

Quality impact: This increases maintenance effort but may be intentional project/course coverage.

Recommendation/status: Keep for this assignment and classify the duplication as context dependent.

5. Pylint Configuration

A project-specific `.pylintrc` was added rather than relying only on defaults. The configuration is intended to make analysis more project-aware, not to manufacture a higher score.

Setting	Default/baseline	Decision	Reason
<code>extension-pkg-allow-list=PySide6</code>	No package allow-list	Change	PySide6's C-extension-backed API can produce false import/member finding
<code>disable=protected-access</code>	Enabled	Change	Tests intentionally inspect private GUI/model methods; this is documented
<code>good-names=tp</code>	Project-specific short name not accepted	Change	tp represents the API field type in the model; avoids artificial renaming solely
<code>max-line-length=100</code>	100	Retain	Retains the standard project limit rather than relaxing it to inflate the score.

The configuration does not disable broad categories such as all warnings/refactoring messages.

6. Implemented Quality Improvements

Improvement	Commit	Quality rationale	Verification
Move MainWindow state initialization	2c5022d	Makes object lifecycle explicit; addresses W0201.	compileall PASS; backend tests PASS
Decompose question transformation	f25825a	Separates extraction, answer construction, scoring and object creation	unittest backend PASS
Reduce repeated parameter-frame set	f25825a	Removes repeated GUI construction logic while preserving	compileall PASS.
Explicit UTF-8 stylesheet decoding	2c5022d	Improves cross-platform reliability; addresses W1514.	compileall PASS
Simplify raise control flow / improve	2c5022d	Addresses R1720, W0622 and C0123 without redesigning	compileall PASS; backend tests PASS

7. Verification

Static syntax: python -m compileall -q main.py modules tests — PASS.

Backend regression: python -m unittest discover -s tests/unittest_backend -p 'test_*.py' — 19 tests ran, 19 passed.

Full pytest/GUI: blocked because PySide6, pytest-mock and pytest-qt are not installed. Collection stopped at pytest_mock import.

Application launch: blocked because PySide6 is unavailable.

8. Before/After Pylint Comparison

Metric	Before	After
Pylint score	4.49/10	Pending local run
Total findings	695	Pending local run
Errors	90	Pending local run
Warnings	323	Pending local run
Refactoring	26	Pending local run
Convention	256	Pending local run

This is the only required evidence item that still needs local execution. The repository contains `deliverable/run_final_analysis.ps1`, which produces both text and JSON output.

```
powershell -ExecutionPolicy Bypass -File deliverable/run_final_analysis.ps1
```

The resulting `pylint_final.txt` and `pylint_final.json` should be added to the submission after the command succeeds. The numerical result must not be invented.

9. Static Analysis vs Human Review

What Pylint detected well: object initialization patterns, broad exception handling, unnecessary else-after-raise, encoding ambiguity, naming conflicts, type-check style, constructor/attribute counts and duplicated test code.

What required human judgment: protected access in tests, pytest fixture naming, GUI classes with many attributes, data-object constructors with many parameters, deliberate duplicate test implementations, and Qt framework-required names such as closeEvent.

Context-dependent findings: PySide6 import/member findings can arise when the analysis environment cannot fully infer the package; protected-access warnings in tests can be intentional; too-few-public-methods can be normal for widgets/value-like classes.

Can a high Pylint score still hide poor design? Yes. A score reflects enabled static rules and weighted message counts; it cannot independently determine cohesion, architectural responsibility, coupling or whether duplication is intentional.

Conclusion: Static analysis should complement human review. Pylint provides repeatable detection of patterns; human review provides architectural context, framework awareness and trade-off analysis.

10. Git History

```
5449e69 (HEAD -> static-analysis) update project selection evidence
ea96954 configure pylint and preserve baseline evidence
2c5022d improve static-analysis findings
f25825a refactor question data transformation
91c1f2e docs: record baseline pylint analysis
ac648a2 (origin/static-analysis) Initial Commit
e4aa48e (origin/main, origin/HEAD, main) docs: update README
d04d13e docs: update README and coverage setup
```

The improvement work is split into meaningful commits rather than one final bulk commit. The branch remains static-analysis.

Appendix A — Complete Initial Pylint Output

The complete baseline output from deliverable/pylint_baseline.txt is included below.

```
***** Module main
main.py:51:0: C0303: Trailing whitespace (trailing-whitespace)
main.py:92:0: C0303: Trailing whitespace (trailing-whitespace)
main.py:1:0: C0114: Missing module docstring (missing-module-docstring)
main.py:6:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
main.py:6:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
main.py:7:0: E0611: No name 'QFont' in module 'PySide6.QtGui' (no-name-in-module)
main.py:13:0: C0116: Missing function or method docstring (missing-function-docstring)
main.py:102:9: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
***** Module modules.gui.main_window
modules\gui\main_window.py:6:0: C0301: Line too long (110/100) (line-too-long)
modules\gui\main_window.py:47:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\main_window.py:162:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\main_window.py:166:0: C0301: Line too long (109/100) (line-too-long)
modules\gui\main_window.py:220:0: C0301: Line too long (104/100) (line-too-long)
modules\gui\main_window.py:224:26: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\main_window.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\main_window.py:3:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\main_window.py:3:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\main_window.py:4:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\main_window.py:14:0: R0902: Too many instance attributes (11/7) (too-many-instance-attributes)
modules\gui\main_window.py:112:15: W0718: Catching too general exception Exception (broad-exception-caught)
modules\gui\main_window.py:209:4: C0103: Method name "closeEvent" doesn't conform to snake_case naming style (invalid-name)
modules\gui\main_window.py:221:12: W0201: Attribute '_close_requested' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:106:12: W0201: Attribute '_question_loader' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:150:8: W0201: Attribute '_question_loader' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:107:12: W0201: Attribute '_thread_controller' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:151:8: W0201: Attribute '_thread_controller' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:86:8: W0201: Attribute '_start_display' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:163:8: W0201: Attribute '_question_display' defined outside __init__ (attribute-defined-outside-init)
modules\gui\main_window.py:14:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.gui
modules\gui\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.menu_bar.menu_actions
modules\gui\menu_bar\menu_actions.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.menu_bar.menu_bar
modules\gui\menu_bar\menu_bar.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\menu_bar\menu_bar.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\menu_bar\menu_bar.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\menu_bar\menu_bar.py:2:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\menu_bar\menu_bar.py:7:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.menu_bar.__init__
modules\gui\menu_bar\__init__.py:2:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.menu_bar
modules\gui\menu_bar\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.menu_bar.menus.help_menu
modules\gui\menu_bar\menus\help_menu.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\menu_bar\menus\help_menu.py:3:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\menu_bar\menus\help_menu.py:3:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\menu_bar\menus\help_menu.py:4:0: E0611: No name 'QAction' in module 'PySide6.QtGui' (no-name-in-module)
modules\gui\menu_bar\menus\help_menu.py:5:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\menu_bar\menus\help_menu.py:5:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\menu_bar\menus\help_menu.py:12:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.menu_bar.menus.quiz_menu
modules\gui\menu_bar\menus\quiz_menu.py:42:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\menu_bar\menus\quiz_menu.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\menu_bar\menus\quiz_menu.py:3:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\menu_bar\menus\quiz_menu.py:3:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\menu_bar\menus\quiz_menu.py:4:0: E0611: No name 'QAction' in module 'PySide6.QtGui' (no-name-in-module)
modules\gui\menu_bar\menus\quiz_menu.py:5:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\menu_bar\menus\quiz_menu.py:5:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\menu_bar\menus\quiz_menu.py:38:8: W0622: Redefining built-in 'exit' (redefined-builtin)
modules\gui\menu_bar\menus\quiz_menu.py:12:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.menu_bar.menus
modules\gui\menu_bar\menus\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.welcome_label
modules\gui\widgets\welcome_label.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\welcome_label.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\welcome_label.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\welcome_label.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\welcome_label.py:5:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.widgets.__init__
modules\gui\widgets\__init__.py:4:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets
modules\gui\widgets\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.overlays.error_overlay
modules\gui\widgets\overlays\error_overlay.py:12:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\overlays\error_overlay.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\overlays\error_overlay.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\overlays\error_overlay.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\overlays\error_overlay.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\overlays\error_overlay.py:2:0: E0611: No name 'QTimer' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\overlays\error_overlay.py:62:11: C0123: Use isinstance() rather than type() for a typecheck. (unidiomatic-typecheck)
modules\gui\widgets\overlays\error_overlay.py:7:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.gui.widgets.overlays.loading_overlay
modules\gui\widgets\overlays\loading_overlay.py:1:0: C0114: Missing module docstring (missing-module-docstring)
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modules\gui\widgets\overlays\loading_overlay.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\overlays\loading_overlay.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\overlays\loading_overlay.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\overlays\loading_overlay.py:43:4: C0116: Missing function or method docstring (missing-function-docstring)
modules\gui\widgets\overlays\loading_overlay.py:47:4: C0116: Missing function or method docstring (missing-function-docstring)
***** Module modules.gui.widgets.overlays.__init__
modules\gui\widgets\overlays\__init__.py:2:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets.overlays
modules\gui\widgets\overlays\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.question_display.question_display
modules\gui\widgets\question_display\question_display.py:67:0: C0301: Line too long (108/100) (line-too-long)
modules\gui\widgets\question_display\question_display.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\question_display\question_display.py:3:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\question_display\question_display.py:3:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\question_display\question_display.py:4:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\question_display\question_display.py:4:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\question_display\question_display.py:4:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\question_display\question_display.py:12:0: R0902: Too many instance attributes (12/7) (too-many-instance-attributes)
modules\gui\widgets\question_display\question_display.py:91:8: W0201: Attribute '_result_widget' defined outside __init__ (attribute-defined-outside-init)
modules\gui\widgets\question_display\question_display.py:100:8: W0201: Attribute '_repeat_quiz_button' defined outside __init__ (attribute-defined-outside-init)
modules\gui\widgets\question_display\question_display.py:12:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.widgets.question_display.__init__
modules\gui\widgets\question_display\__init__.py:1:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets.question_display
modules\gui\widgets\question_display\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.question_display.components.question_widget
modules\gui\widgets\question_display\components\question_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\question_display\components\question_widget.py:5:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\question_display\components\question_widget.py:5:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\question_display\components\question_widget.py:6:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\question_display\components\question_widget.py:91:4: C0116: Missing function or method docstring (missing-function-docstring)
modules\gui\widgets\question_display\components\question_widget.py:95:4: C0116: Missing function or method docstring (missing-function-docstring)
***** Module modules.gui.widgets.question_display.components.result_widget
modules\gui\widgets\question_display\components\result_widget.py:9:17: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:10:33: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:11:39: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:12:29: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:28:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:39:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\question_display\components\result_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\question_display\components\result_widget.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\question_display\components\result_widget.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\question_display\components\result_widget.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\question_display\components\result_widget.py:5:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.widgets.question_display.components.__init__
modules\gui\widgets\question_display\components\__init__.py:2:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets.question_display.components
modules\gui\widgets\question_display\components\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.start_display.start_display
modules\gui\widgets\start_display\start_display.py:6:0: C0301: Line too long (115/100) (line-too-long)
modules\gui\widgets\start_display\start_display.py:45:0: C0301: Line too long (103/100) (line-too-long)
modules\gui\widgets\start_display\start_display.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\start_display\start_display.py:3:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\start_display\start_display.py:3:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\start_display\start_display.py:4:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\start_display.py:4:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\start_display.py:4:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\start_display.py:12:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.widgets.start_display.__init__
modules\gui\widgets\start_display\__init__.py:2:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets.start_display
modules\gui\widgets\start_display\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.widgets.start_display.components.app_name_widget
modules\gui\widgets\start_display\components\app_name_widget.py:22:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\start_display\components\app_name_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\start_display\components\app_name_widget.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\start_display\components\app_name_widget.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\start_display\components\app_name_widget.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\app_name_widget.py:5:0: R0903: Too few public methods (0/2) (too-few-public-methods)
***** Module modules.gui.widgets.start_display.components.question_params_widget
modules\gui\widgets\start_display\components\question_params_widget.py:73:82: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\start_display\components\question_params_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\start_display\components\question_params_widget.py:5:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\start_display\components\question_params_widget.py:5:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\start_display\components\question_params_widget.py:6:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\question_params_widget.py:6:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\question_params_widget.py:11:0: C0115: Missing class docstring (missing-class-docstring)
modules\gui\widgets\start_display\components\question_params_widget.py:49:4: C0116: Missing function or method docstring (missing-function-docstring)
modules\gui\widgets\start_display\components\question_params_widget.py:27:0: R0903: Too few public methods (1/2) (too-few-public-methods)
modules\gui\widgets\start_display\components\question_params_widget.py:56:0: R0902: Too many instance attributes (10/7) (too-many-instance-attributes)
***** Module modules.gui.widgets.start_display.components.start_button_widget
modules\gui\widgets\start_display\components\start_button_widget.py:24:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\widgets\start_display\components\start_button_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\widgets\start_display\components\start_button_widget.py:1:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
modules\gui\widgets\start_display\components\start_button_widget.py:1:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
modules\gui\widgets\start_display\components\start_button_widget.py:2:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\start_button_widget.py:2:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\start_button_widget.py:2:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\widgets\start_display\components\start_button_widget.py:35:4: C0116: Missing function or method docstring (missing-function-docstring)
modules\gui\widgets\start_display\components\start_button_widget.py:5:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.gui.widgets.start_display.components.__init__

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modules\gui\widgets\start_display\components\__init__.py:3:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.widgets.start_display.components
modules\gui\widgets\start_display\components\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.gui.workers.question_loader
modules\gui\workers\question_loader.py:48:0: C0301: Line too long (105/100) (line-too-long)
modules\gui\workers\question_loader.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\workers\question_loader.py:3:0: E0611: No name 'QObject' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\question_loader.py:3:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\question_loader.py:3:0: E0611: No name 'Slot' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\question_loader.py:51:15: W0718: Catching too general exception Exception (broad-exception-caught)
modules\gui\workers\question_loader.py:11:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.gui.workers.worker_thread_controller
modules\gui\workers\worker_thread_controller.py:83:0: C0303: Trailing whitespace (trailing-whitespace)
modules\gui\workers\worker_thread_controller.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\gui\workers\worker_thread_controller.py:7:0: E0611: No name 'QObject' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\worker_thread_controller.py:7:0: E0611: No name 'Signal' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\worker_thread_controller.py:7:0: E0611: No name 'QThread' in module 'PySide6.QtCore' (no-name-in-module)
modules\gui\workers\worker_thread_controller.py:71:4: C0116: Missing function or method docstring (missing-function-docstring)
modules\gui\workers\worker_thread_controller.py:76:15: W0718: Catching too general exception Exception (broad-exception-caught)
modules\gui\workers\worker_thread_controller.py:81:4: C0116: Missing function or method docstring (missing-function-docstring)
***** Module modules.gui.workers.__init__
modules\gui\workers\__init__.py:2:0: C0304: Final newline missing (missing-final-newline)
***** Module modules.gui.workers
modules\gui\workers\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module modules.questions.models
modules\questions\models.py:12:0: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:81:0: C0301: Line too long (103/100) (line-too-long)
modules\questions\models.py:85:47: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:92:0: C0301: Line too long (106/100) (line-too-long)
modules\questions\models.py:114:19: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:115:27: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:116:25: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:117:25: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:118:31: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:119:34: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:120:28: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:122:0: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\models.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\questions\models.py:10:0: R0902: Too many instance attributes (8/7) (too-many-instance-attributes)
modules\questions\models.py:13:4: R0913: Too many arguments (8/5) (too-many-arguments)
modules\questions\models.py:13:4: R0917: Too many positional arguments (8/5) (too-many-positional-arguments)
modules\questions\models.py:10:0: R0903: Too few public methods (0/2) (too-few-public-methods)
modules\questions\models.py:33:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.questions.opentdb_client
modules\questions\opentdb_client.py:83:0: C0301: Line too long (134/100) (line-too-long)
modules\questions\opentdb_client.py:85:0: C0301: Line too long (122/100) (line-too-long)
modules\questions\opentdb_client.py:92:0: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\opentdb_client.py:107:0: C0303: Trailing whitespace (trailing-whitespace)
modules\questions\opentdb_client.py:113:0: C0304: Final newline missing (missing-final-newline)
modules\questions\opentdb_client.py:1:0: C0114: Missing module docstring (missing-module-docstring)
modules\questions\opentdb_client.py:21:0: C0115: Missing class docstring (missing-class-docstring)
modules\questions\opentdb_client.py:30:0: C0115: Missing class docstring (missing-class-docstring)
modules\questions\opentdb_client.py:82:12: R1720: Unnecessary "elif" after "raise", remove the leading "el" from "elif" (no-else-raise)
modules\questions\opentdb_client.py:35:0: R0903: Too few public methods (1/2) (too-few-public-methods)
***** Module modules.questions.__init__
modules\questions\__init__.py:3:0: C0301: Line too long (126/100) (line-too-long)
***** Module modules.questions
modules\questions\__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
***** Module tests.data.api_response_test_data
tests\data\api_response_test_data.py:48:0: C0301: Line too long (122/100) (line-too-long)
tests\data\api_response_test_data.py:127:0: C0304: Final newline missing (missing-final-newline)
***** Module tests.gui.test_main_window
tests\gui\test_main_window.py:46:0: C0301: Line too long (106/100) (line-too-long)
tests\gui\test_main_window.py:61:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:62:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:93:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:94:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:112:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:113:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:133:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:134:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:268:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:269:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:287:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:288:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:306:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:307:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:336:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:337:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:364:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:365:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:384:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:385:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:407:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:408:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:427:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:436:36: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:456:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:457:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:502:0: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_main_window.py:1:0: C0114: Missing module docstring (missing-module-docstring)

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[illegible]

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tests\gui\test_main_window.py:450:4: W0212: Access to a protected member _clear_window of a client class (protected-access)
tests\gui\test_main_window.py:452:11: W0212: Access to a protected member _main_layout of a client class (protected-access)
tests\gui\test_main_window.py:458:8: W0621: Redefining name 'main_window' from outer scope (line 17) (redefined-outer-name)
tests\gui\test_main_window.py:473:37: W0212: Access to a protected member _main_layout of a client class (protected-access)
tests\gui\test_main_window.py:474:45: W0212: Access to a protected member _main_layout of a client class (protected-access)
tests\gui\test_main_window.py:476:4: W0212: Access to a protected member _display_widget of a client class (protected-access)
tests\gui\test_main_window.py:488:8: W0621: Redefining name 'main_window' from outer scope (line 17) (redefined-outer-name)
tests\gui\test_main_window.py:501:4: W0212: Access to a protected member _thread_controller of a client class (protected-access)
tests\gui\test_main_window.py:508:11: W0212: Access to a protected member _close_requested of a client class (protected-access)
tests\gui\test_main_window.py:516:8: W0621: Redefining name 'main_window' from outer scope (line 17) (redefined-outer-name)
tests\gui\test_main_window.py:530:4: W0212: Access to a protected member _thread_controller of a client class (protected-access)
tests\gui\test_main_window.py:537:11: W0212: Access to a protected member _close_requested of a client class (protected-access)
***** Module tests.gui.test_question_display
tests\gui\test_question_display.py:18:0: C0301: Line too long (106/100) (line-too-long)
tests\gui\test_question_display.py:55:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:56:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:73:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:74:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:86:7: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:106:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:164:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_display.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\gui\test_question_display.py:2:0: E0401: Unable to import 'pytest_mock' (import-error)
tests\gui\test_question_display.py:3:0: E0401: Unable to import 'pytestqt.qtbot' (import-error)
tests\gui\test_question_display.py:17:4: W0621: Redefining name 'questions' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_display.py:23:19: W0621: Redefining name 'questions' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_display.py:25:4: W0621: Redefining name 'questions_list' from outer scope (line 23) (redefined-outer-name)
tests\gui\test_question_display.py:29:21: W0621: Redefining name 'questions_list' from outer scope (line 23) (redefined-outer-name)
tests\gui\test_question_display.py:34:4: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:38:35: W0621: Redefining name 'questions_list' from outer scope (line 23) (redefined-outer-name)
tests\gui\test_question_display.py:43:4: W0621: Redefining name 'question_widget_list_formatted' from outer scope (line 38) (redefined-outer-name)
tests\gui\test_question_display.py:47:12: W0212: Access to a protected member _correct_answer of a client class (protected-access)
tests\gui\test_question_display.py:49:12: W0212: Access to a protected member _correct_answer of a client class (protected-access)
tests\gui\test_question_display.py:57:8: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:68:4: W0212: Access to a protected member _finish_quiz_button of a client class (protected-access)
tests\gui\test_question_display.py:75:8: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:93:4: W0212: Access to a protected member _on_finish_quiz_button_clicked of a client class (protected-access)
tests\gui\test_question_display.py:95:11: W0212: Access to a protected member _finish_quiz_button of a client class (protected-access)
tests\gui\test_question_display.py:108:8: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:118:18: W0212: Access to a protected member _widget_list of a client class (protected-access)
tests\gui\test_question_display.py:122:4: W0212: Access to a protected member _calculate_result_and_format_widgets of a client class (protected-access)
tests\gui\test_question_display.py:129:8: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:130:8: W0621: Redefining name 'question_widget_list_formatted' from outer scope (line 38) (redefined-outer-name)
tests\gui\test_question_display.py:149:4: W0212: Access to a protected member _widget_list of a client class (protected-access)
tests\gui\test_question_display.py:152:34: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_display.py:153:11: W0212: Access to a protected member _correct_answer of a client class (protected-access)
tests\gui\test_question_display.py:155:40: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_display.py:157:4: W0212: Access to a protected member _calculate_result_and_format_widgets of a client class (protected-access)
tests\gui\test_question_display.py:159:11: W0212: Access to a protected member _questions_total_points of a client class (protected-access)
tests\gui\test_question_display.py:160:11: W0212: Access to a protected member _user_correct_answer_amount of a client class (protected-access)
tests\gui\test_question_display.py:161:11: W0212: Access to a protected member _user_total_earned_points of a client class (protected-access)
tests\gui\test_question_display.py:165:8: W0621: Redefining name 'question_display' from outer scope (line 29) (redefined-outer-name)
tests\gui\test_question_display.py:172:4: W0212: Access to a protected member _on_finish_quiz_button_clicked of a client class (protected-access)
tests\gui\test_question_display.py:174:20: W0212: Access to a protected member _repeat_quiz_button of a client class (protected-access)
tests\gui\test_question_display.py:7:0: C0412: Imports from package modules are not grouped (ungrouped-imports)
***** Module tests.gui.test_question_loader
tests\gui\test_question_loader.py:34:18: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:41:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:76:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:77:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:103:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:104:30: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_loader.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\gui\test_question_loader.py:2:0: E0401: Unable to import 'pytest_mock' (import-error)
tests\gui\test_question_loader.py:3:0: E0401: Unable to import 'pytestqt.qtbot' (import-error)
tests\gui\test_question_loader.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_question_loader.py:12:4: W0621: Redefining name 'question_params' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_loader.py:21:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_question_loader.py:21:20: W0621: Redefining name 'question_params' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_loader.py:22:4: W0621: Redefining name 'question_loader' from outer scope (line 21) (redefined-outer-name)
tests\gui\test_question_loader.py:32:4: W0621: Redefining name 'questions' from outer scope (line 26) (redefined-outer-name)
tests\gui\test_question_loader.py:42:8: W0621: Redefining name 'question_params' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_loader.py:43:8: W0621: Redefining name 'questions' from outer scope (line 26) (redefined-outer-name)
tests\gui\test_question_loader.py:58:4: W0621: Redefining name 'question_loader' from outer scope (line 21) (redefined-outer-name)
tests\gui\test_question_loader.py:78:8: W0621: Redefining name 'question_params' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_loader.py:79:8: W0621: Redefining name 'questions' from outer scope (line 26) (redefined-outer-name)
tests\gui\test_question_loader.py:86:4: W0621: Redefining name 'question_loader' from outer scope (line 21) (redefined-outer-name)
tests\gui\test_question_loader.py:105:8: W0621: Redefining name 'question_params' from outer scope (line 11) (redefined-outer-name)
tests\gui\test_question_loader.py:112:4: W0621: Redefining name 'question_loader' from outer scope (line 21) (redefined-outer-name)
***** Module tests.gui.test_question_params
tests\gui\test_question_params.py:11:0: C0301: Line too long (137/100) (line-too-long)
tests\gui\test_question_params.py:27:12: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:32:12: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:39:47: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:81:0: C0301: Line too long (113/100) (line-too-long)
tests\gui\test_question_params.py:92:0: C0301: Line too long (114/100) (line-too-long)
tests\gui\test_question_params.py:120:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:163:21: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:164:53: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:180:17: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_question_params.py:200:21: C0303: Trailing whitespace (trailing-whitespace)

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tests\gui\test_question_widget.py:5:0: E0401: Unable to import 'pytestqt.qtbot' (import-error)
tests\gui\test_question_widget.py:18:4: W0621: Redefining name 'questions' from outer scope (line 12) (redefined-outer-name)
tests\gui\test_question_widget.py:24:13: W0621: Redefining name 'questions' from outer scope (line 12) (redefined-outer-name)
tests\gui\test_question_widget.py:26:4: W0621: Redefining name 'question' from outer scope (line 24) (redefined-outer-name)
tests\gui\test_question_widget.py:30:20: W0621: Redefining name 'question' from outer scope (line 24) (redefined-outer-name)
tests\gui\test_question_widget.py:51:25: W0621: Redefining name 'questions' from outer scope (line 12) (redefined-outer-name)
tests\gui\test_question_widget.py:53:4: W0621: Redefining name 'question_widget_list' from outer scope (line 51) (redefined-outer-name)
tests\gui\test_question_widget.py:54:11: W0621: Redefining name 'question' from outer scope (line 24) (redefined-outer-name)
tests\gui\test_question_widget.py:55:8: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:59:50: W0621: Redefining name 'question_widget_list' from outer scope (line 51) (redefined-outer-name)
tests\gui\test_question_widget.py:66:11: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:69:15: W0212: Access to a protected member _label of a client class (protected-access)
tests\gui\test_question_widget.py:71:61: W0621: Redefining name 'question_widget_list' from outer scope (line 51) (redefined-outer-name)
tests\gui\test_question_widget.py:79:8: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:83:24: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:84:22: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:85:23: W0212: Access to a protected member _answer_button_group of a client class (protected-access)
tests\gui\test_question_widget.py:97:71: W0621: Redefining name 'question_widget_list' from outer scope (line 51) (redefined-outer-name)
tests\gui\test_question_widget.py:106:8: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:111:23: W0212: Access to a protected member _answer_button_group of a client class (protected-access)
tests\gui\test_question_widget.py:112:25: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:123:19: W0212: Access to a protected member _correct_answer of a client class (protected-access)
tests\gui\test_question_widget.py:125:60: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:133:18: W0212: Access to a protected member _answer_button_group of a client class (protected-access)
tests\gui\test_question_widget.py:135:25: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:140:4: W0212: Access to a protected member _color_widget_frame_on_finish of a client class (protected-access)
tests\gui\test_question_widget.py:145:62: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:154:4: W0212: Access to a protected member _color_widget_frame_on_finish of a client class (protected-access)
tests\gui\test_question_widget.py:159:18: W0212: Access to a protected member _answer_button_group of a client class (protected-access)
tests\gui\test_question_widget.py:161:25: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:166:4: W0212: Access to a protected member _color_widget_frame_on_finish of a client class (protected-access)
tests\gui\test_question_widget.py:171:48: W0621: Redefining name 'question_widget_list' from outer scope (line 51) (redefined-outer-name)
tests\gui\test_question_widget.py:178:8: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:181:23: W0212: Access to a protected member _answer_button_group of a client class (protected-access)
tests\gui\test_question_widget.py:184:8: W0212: Access to a protected member _format_buttons_on_finish of a client class (protected-access)
tests\gui\test_question_widget.py:189:29: W0212: Access to a protected member _question of a client class (protected-access)
tests\gui\test_question_widget.py:198:67: W0621: Redefining name 'question_widget' from outer scope (line 30) (redefined-outer-name)
tests\gui\test_question_widget.py:9:0: C0412: Imports from package modules are not grouped (ungrouped-imports)
***** Module tests.gui.test_start_button_widget
tests\gui\test_start_button_widget.py:13:0: C0301: Line too long (116/100) (line-too-long)
tests\gui\test_start_button_widget.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\gui\test_start_button_widget.py:2:0: E0401: Unable to import 'pytestqt.qtbot' (import-error)
tests\gui\test_start_button_widget.py:4:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
tests\gui\test_start_button_widget.py:5:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
tests\gui\test_start_button_widget.py:5:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
tests\gui\test_start_button_widget.py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_start_button_widget.py:13:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_start_button_widget.py:13:75: W0621: Redefining name 'start_button' from outer scope (line 10) (redefined-outer-name)
tests\gui\test_start_button_widget.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_start_button_widget.py:24:54: W0621: Redefining name 'start_button' from outer scope (line 10) (redefined-outer-name)
tests\gui\test_start_button_widget.py:4:0: W0611: Unused Qt imported from PySide6.QtCore (unused-import)
***** Module tests.gui.test_start_display
tests\gui\test_start_display.py:74:22: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_start_display.py:79:36: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_start_display.py:84:39: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_start_display.py:114:39: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_start_display.py:141:39: C0303: Trailing whitespace (trailing-whitespace)
tests\gui\test_start_display.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\gui\test_start_display.py:2:0: E0401: Unable to import 'pytest_mock' (import-error)
tests\gui\test_start_display.py:3:0: E0401: Unable to import 'pytestqt.qtbot' (import-error)
tests\gui\test_start_display.py:5:0: E0611: No name 'Qt' in module 'PySide6.QtCore' (no-name-in-module)
tests\gui\test_start_display.py:6:0: E0401: Unable to import 'PySide6.QtWidgets' (import-error)
tests\gui\test_start_display.py:6:0: E0611: No name 'QtWidgets' in module 'PySide6' (no-name-in-module)
tests\gui\test_start_display.py:13:0: C0116: Missing function or method docstring (missing-function-docstring)
tests\gui\test_start_display.py:13:18: W0613: Unused argument 'mock' (unused-argument)
tests\gui\test_start_display.py:19:8: W0621: Redefining name 'start_display' from outer scope (line 13) (redefined-outer-name)
tests\gui\test_start_display.py:32:26: W0212: Access to a protected member _start_button of a client class (protected-access)
tests\gui\test_start_display.py:39:8: W0621: Redefining name 'start_display' from outer scope (line 13) (redefined-outer-name)
tests\gui\test_start_display.py:54:22: W0212: Access to a protected member _question_params of a client class (protected-access)
tests\gui\test_start_display.py:38:8: W0613: Unused argument 'mock' (unused-argument)
tests\gui\test_start_display.py:61:8: W0621: Redefining name 'start_display' from outer scope (line 13) (redefined-outer-name)
tests\gui\test_start_display.py:79:8: W0212: Access to a protected member _start_button of a client class (protected-access)
tests\gui\test_start_display.py:84:8: W0212: Access to a protected member _question_params of a client class (protected-access)
tests\gui\test_start_display.py:99:8: W0621: Redefining name 'start_display' from outer scope (line 13) (redefined-outer-name)
tests\gui\test_start_display.py:114:8: W0212: Access to a protected member _question_params of a client class (protected-access)
tests\gui\test_start_display.py:127:8: W0621: Redefining name 'start_display' from outer scope (line 13) (redefined-outer-name)
tests\gui\test_start_display.py:141:8: W0212: Access to a protected member _question_params of a client class (protected-access)
tests\gui\test_start_display.py:5:0: W0611: Unused Qt imported from PySide6.QtCore (unused-import)
***** Module tests.pytest_backend.test_models_pytest
tests\pytest_backend\test_models_pytest.py:6:9: C0303: Trailing whitespace (trailing-whitespace)
tests\pytest_backend\test_models_pytest.py:32:0: C0301: Line too long (101/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:33:0: C0301: Line too long (103/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:41:0: C0301: Line too long (144/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:44:0: C0301: Line too long (103/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:53:0: C0301: Line too long (109/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:56:0: C0301: Line too long (103/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:64:0: C0301: Line too long (125/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:70:0: C0301: Line too long (111/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:77:0: C0301: Line too long (162/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:80:0: C0301: Line too long (134/100) (line-too-long)
tests\pytest_backend\test_models_pytest.py:82:0: C0301: Line too long (147/100) (line-too-long)

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[illegible]

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tests\unittest_backend\test_models_unittest.py:75:44: C0303: Trailing whitespace (trailing-whitespace)
tests\unittest_backend\test_models_unittest.py:137:0: C0303: Trailing whitespace (trailing-whitespace)
tests\unittest_backend\test_models_unittest.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\unittest_backend\test_models_unittest.py:4:0: C0410: Multiple imports on one line (html, re, random) (multiple-imports)
tests\unittest_backend\test_models_unittest.py:11:0: C0115: Missing class docstring (missing-class-docstring)
tests\unittest_backend\test_models_unittest.py:39:25: W0212: Access to a protected member _get_question_data_from_api_client of a client
tests\unittest_backend\test_models_unittest.py:52:12: W0212: Access to a protected member _get_question_data_from_api_client of a client
tests\unittest_backend\test_models_unittest.py:60:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
tests\unittest_backend\test_models_unittest.py:84:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
tests\unittest_backend\test_models_unittest.py:96:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
tests\unittest_backend\test_models_unittest.py:107:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
tests\unittest_backend\test_models_unittest.py:111:16: W0612: Unused variable 'q_key' (unused-variable)
tests\unittest_backend\test_models_unittest.py:128:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
tests\unittest_backend\test_models_unittest.py:143:8: W0212: Access to a protected member _questions_data_to_question_objects of a client
***** Module tests.unittest_backend.test_opentdb_client_unittest
tests\unittest_backend\test_opentdb_client_unittest.py:17:0: C0301: Line too long (104/100) (line-too-long)
tests\unittest_backend\test_opentdb_client_unittest.py:85:0: C0301: Line too long (107/100) (line-too-long)
tests\unittest_backend\test_opentdb_client_unittest.py:100:0: C0301: Line too long (101/100) (line-too-long)
tests\unittest_backend\test_opentdb_client_unittest.py:104:0: C0301: Line too long (115/100) (line-too-long)
tests\unittest_backend\test_opentdb_client_unittest.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tests\unittest_backend\test_opentdb_client_unittest.py:9:0: C0115: Missing class docstring (missing-class-docstring)
tests\unittest_backend\test_opentdb_client_unittest.py:1:0: R0801: Similar lines in 2 files
==tests.gui.test_main_window: [51:73]
==tests.gui.test_question_loader: [11:21]
    question_params: QuestionParams = {
        "amount": "10",
        "difficulty": "",
        "category": "",
        "question_type": "",
    }
    return question_params

@pytest.fixture
def question_loader(question_params) -> QuestionLoader: (duplicate-code)
tests\unittest_backend\test_opentdb_client_unittest.py:1:0: R0801: Similar lines in 2 files
==tests.pytest_backend.test_opentdb_client_pytest: [62:73]
==tests.unittest_backend.test_opentdb_client_unittest: [74:86]
    "category": "9",
    "difficulty": "easy",
    "type": "boolean",
    },
    timeout=3,
)

    mock_response.raise_for_status.assert_called_once()

    @patch("modules.questions.opentdb_client.requests.get")
    def test_get_questions_data_raises_error_when_no_questions(self, mock_requests_get: MagicMock) -> None:
        """Test that an empty results list raises OpenTriviaClientError.""" (duplicate-code)
tests\unittest_backend\test_opentdb_client_unittest.py:1:0: R0801: Similar lines in 2 files
==tests.pytest_backend.test_opentdb_client_pytest: [33:43]
==tests.unittest_backend.test_opentdb_client_unittest: [38:50]
    mock_requests_get.assert_called_once_with(
        OpenTriviaClient.BASE_URL,
        params={"amount": 1},
        timeout=3,
    )

    # Check whether raise_for_status has been called on mocked response
    mock_response.raise_for_status.assert_called_once()

    @patch("modules.questions.opentdb_client.requests.get")
    def test_success_get_questions_data_with_all_params(self, mock_requests_get: MagicMock) -> None:
        """Test successful data retrieval with all request parameters.""" (duplicate-code)

-----
Your code has been rated at 4.49/10

```